

How Does Artificial Intelligence Influence Critical Success Factors in Software Development Projects?

A Scoped Systematic Literature Review

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Abstract

Software development projects continue to struggle to meet agreed schedules, budgets, quality levels and stakeholder expectations. Artificial intelligence (AI) is increasingly applied to planning, forecasting, monitoring, coding, testing and decision support, yet few studies systematically relate this evidence to established critical success factors. This paper asks how AI adoption influences critical success factors and project success in software development projects. It follows a systematic literature review protocol: a scoped database search, supplemented by snowballing. The review uses the taxonomy of 50 software-project critical success factors from Monserrat et al. (2025) as a baseline for mapping AI applications. The synthesis indicates that AI is not an independent or universal success factor. Instead, AI alters the effectiveness of established factors such as project planning, schedule management, economic management, risk management, quality management, communication and knowledge management. Its contribution appears contingent on data quality, governance, organisational readiness, participant skills, explainability, privacy protection and human oversight. AI seems most useful in structured, data-intensive activities, including forecasting, anomaly detection, resource allocation, coding support, documentation and testing. It also introduces risks: bias, opacity, security, intellectual property, overreliance and weak stakeholder trust. The paper concludes that AI acts as a conditional amplifier of traditional critical success factors, not a technology that automatically improves project outcomes; empirical evidence linking specific AI applications to measurable project outcomes remains limited.

Keywords: artificial intelligence; software project management; critical success factors; project success; systematic literature review

1 Introduction

Software development organisations are predominantly project-based, yet project failure and underperformance remain persistent concerns. Traditional project-management research has

often assessed success through the *iron triangle*, that is, delivering within the planned schedule and budget while meeting agreed scope and quality requirements. This definition is increasingly regarded as incomplete, because a project can meet its technical delivery targets while failing to achieve user acceptance, stakeholder satisfaction, strategic value or durable operational benefits (Ika 2009; McLeod and MacDonell 2011).

Monserrat et al. (2025), citing widely circulated Standish Group figures for 2020, report that 31 per cent of projects were classified as successful, 50 per cent as challenged and 19 per cent as failed. They also show that success extends beyond time, budget and scope to sponsor objectives, customer feedback, product usability and stakeholder expectations. Related work agrees: practitioners weight stakeholder and customer-related criteria heavily when judging whether a project succeeded (Procaccino et al. 2005; Procaccino and Verner 2006). Perceived causes of failure similarly cluster around people, process and requirements rather than technology alone (Lehtinen et al. 2014).

Artificial intelligence (AI) is increasingly relevant in this context. AI-enabled tools support effort estimation, scheduling, risk prediction, monitoring, quality assurance, documentation, code generation, testing and decision-making (Taboada et al. 2023; Adamantiadou and Tsironis 2025). Machine-learning techniques can process historical project data to identify patterns associated with delays, cost overruns, defects or risk (Mahdi et al. 2021). Generative AI tools can help developers and project teams write code, prepare documentation, summarise project information, generate test cases and retrieve organisational knowledge (Russo 2024).

Concluding that AI simply improves project success would nevertheless be premature. The usefulness of these tools depends on data relevance and quality, integration with existing processes, participant competence, user acceptance, and clear accountability and governance. AI can also generate false signals, reproduce historical bias, expose sensitive data, create security or intellectual-property exposure, and encourage overreliance on automated recommendations.

This paper therefore examines AI through the lens of critical success factors (CSFs), the conditions that materially raise the likelihood of project success when addressed properly.

The primary research question is:

RQ: How does the adoption of artificial intelligence influence critical success factors and project success in software development projects?

Two sub-questions guide the analysis:

- **RQ1:** Which established CSFs do AI applications affect, and what contribution to project success does each application potentially offer?
- **RQ2:** Which AI-specific risks and enabling conditions determine whether that effect improves or harms project outcomes?

These two questions bound the review deliberately. The maturity of the underlying evidence and the problem of measuring CSF impact are not posed as separate sub-questions but are taken up where they arise, in Section 4.5 and Section 5.5.

Building on the CSF taxonomy of Monserrat et al. (2025), this paper argues that AI does not constitute an independent, universal CSF. Rather, AI modifies the effectiveness of established

factors, particularly planning, monitoring, risk management, decision quality and knowledge management, while making data governance, organisational readiness, participant skills and human oversight additional conditions for success.

Section 2 reviews project-success criteria, CSF theory and AI in software project management; Section 3 sets out the review protocol; Section 4 maps AI applications onto the CSF baseline; and Section 5 interprets that mapping and derives practical implications.

2 Related Work

2.1 Project Success and Failure: Traditional and Multidimensional Views

Traditional project-management theory evaluates success through the iron triangle of time, cost and scope or quality, under which projects fail when outcomes deviate materially from planned schedules, budgets or specifications. Research has nevertheless criticised this view as too narrow (Ika 2009). A system delivered on time and within budget can still fail if users reject it, if it does not support business objectives, or if it produces no sustained organisational benefit. A project that overruns its schedule, conversely, may still succeed if it creates substantial value or satisfies key stakeholders. McLeod and MacDonell (2011) show that development-project outcomes depend on social, organisational and contextual factors rather than on delivery control alone, and Procaccino et al. (2005) report that practitioners’ own success judgements draw heavily on customer and sponsor satisfaction.

Monserrat et al. (2025) identify the absence of a consolidated definition of software-project success and failure as a central gap, a divergence also visible across three decades of information-systems project-success surveys (Loureiro et al. 2024). Their review shows that some studies rely primarily on iron-triangle criteria, whereas others incorporate sponsor satisfaction, customer feedback, usability and stakeholder expectations. These findings support a multidimensional view of success, combining delivery performance, stakeholder acceptance, product usefulness, strategic alignment and benefits realisation.

The distinction matters for AI-enabled projects. An AI tool may increase delivery speed or improve estimation accuracy without improving user satisfaction, team trust or long-term business value. Judging AI solely by productivity metrics therefore risks overlooking important dimensions of project success – one reason this paper adopts the multidimensional view.

2.2 Critical Success Factors in Software Projects

CSFs are conditions, events or circumstances that significantly increase the likelihood of project success when addressed properly. They must be distinguished from success *criteria*: criteria are the measures used to judge whether a project succeeded, whereas CSFs are the explanatory variables that influence the likelihood of success. The literature frequently blurs this distinction, contributing to the terminological inconsistency discussed below.

Software-engineering research has produced a substantial body of CSF studies. Chow and Cao (2008) surveyed CSFs in agile projects and identified delivery strategy, agile engineering practice and team capability as recurring factors. Ahimbisibwe et al. (2015) argue that which factors matter depends on the chosen methodology, so agile and plan-based projects require

different configurations; methodology choice has also been linked to client satisfaction (Woźniak 2021). Niazi et al. (2016) examine the additional coordination, communication and cultural factors that arise in globally distributed development, and Iriarte and Bayona (2020) consolidate success factors across the wider information-technology project literature. Taken together, these studies indicate that CSFs are numerous, partly context-dependent and rarely reducible to technical capability (Wohlin and Andrews 2003; Almajed and Mayhew 2013; Zaleski and Michalski 2021).

Monserrat et al. (2025) synthesise this body of work. Their systematic literature review searched six information sources – IEEE Xplore, ACM Portal, Springer Link, ScienceDirect, Google Scholar and Wiley Online Library – and initially identified 1,982 studies. After screening, snowballing, full-text analysis and quality assessment, they retained 22 primary studies. The resulting synthesis identifies 50 CSFs grouped into six categories, summarised in Table 1.

Table 1: Categories of software-project critical success factors and representative factors, based on Monserrat et al. (2025).

CSF category	Representative factors
Team and People	Participation, commitment, support and motivation of participants; communication; skills, competencies and capacity; knowledge; training; experience; team characteristics; confidence and security
Organisational	Organisational experience; organisational maturity; organisational environment; policies and rules of the organisation; organisational culture; organisational structure
Technical	Maturity, compatibility and reliability of technology; technical complexity; quality of technology; availability of tools and infrastructure
Development	Specification of project requirements; simple design; coding standards; testing of implemented components; maintenance and evolution of software; software-development life-cycle model
Process and Project Management	Alignment of project objectives with the business model; scope definition; economic management; schedule management; quality management; methodology for managing projects; risk management; change management in project planning; project planning; stakeholder management
Project	Project documentation; project size; complexity; uncertainty; urgency; criticality; security

Of the six categories, *Team and People*, *Organisational* and *Process and Project Management* (Table 1) are the most directly implicated in AI adoption. Team and People factors – skills, knowledge, training, experience and confidence – determine whether participants can understand, use and interrogate AI-generated output. Organisational factors – culture, structure, policies and maturity – determine whether AI can be integrated into working practices at all. Process and Project Management factors identify the specific management activities, from planning and scheduling to quality, risk and change management, that AI applications can support.

Monserrat et al. (2025) also identify limitations in existing CSF research: terminology is inconsistent across studies, many factors lack explicit definitions, and indicators to measure each factor’s actual impact are largely absent. These limitations create a specific opportunity for AI-focused research, because AI may render certain factors more observable and measurable

while introducing new measurement and governance problems.

2.3 Artificial Intelligence in Project and Software Management

AI-enabled project management encompasses machine learning, predictive analytics, optimisation techniques, natural-language processing, intelligent monitoring and generative AI tools used to support project work. In software development, these techniques are applied to planning, estimation, scheduling, risk assessment, quality assurance, defect prediction, documentation, coding, testing and resource allocation (Taboada et al. 2023; Mahdi et al. 2021). Fernandez Rodriguez and Magaña Martínez (2015) reviewed AI applications aimed specifically at project success and factor identification, although that survey predates the generative-AI period.

Machine-learning applications have a long history in software project management, particularly in effort estimation, defect prediction, risk assessment and quality-related analysis (Mahdi et al. 2021; Akimova et al. 2021). These applications map primarily onto the Process and Project Management category: they analyse historical data to estimate effort, identify patterns associated with delivery delays, or surface risks earlier than manual reporting would allow. Generative AI has since widened the practical scope (Russo 2024). Code assistants can produce code fragments, documentation, tests, summaries and explanations, potentially strengthening participant skills, knowledge sharing and delivery speed. Yet generated output may be inaccurate, insecure, difficult to maintain, or legally problematic where intellectual-property questions are ignored.

AI can also support project monitoring and control. Intelligent dashboards can integrate data from issue trackers, version-control systems, test suites, communication platforms and time-tracking tools; they may flag anomalies, predict schedule slippage, detect quality problems or prioritise risks. Decision-support systems help project managers process large volumes of data, but they raise questions of explainability, accountability, automation bias and the appropriate division of responsibility between systems and humans – questions the general literature on explainable decision-support systems treats as central (Kostopoulos et al. 2024).

The literature therefore suggests an uneven pattern. AI shows clear potential in analytical, data-intensive and repetitive activities, whereas its effectiveness is far less certain in leadership-intensive tasks such as conflict resolution, stakeholder negotiation, trust-building and contextual judgement. An expert survey across the PMBOK knowledge areas found the same asymmetry, anticipating a stronger AI influence on cost, schedule and risk management than on human-resource and stakeholder management (Fridgeirsson et al. 2021). This pattern supports a human-augmentation rather than a human-replacement view of AI-enabled project management (Müller et al. 2024), and motivates the mapping exercise in Section 4.

3 Methodology

3.1 Review Design and Research Questions

This paper presents a scoped systematic literature review with supplementary snowballing. It uses a transparent, systematic search protocol to identify relevant research, but the resulting synthesis is narrative and conceptual rather than exhaustive or meta-analytic. The search and

selection protocol follows the established guidelines for SLRs in software engineering (Kitchenham and Charters 2007), complemented by snowballing to identify studies that keyword searches may miss (Wohlin 2014). The approach suits a topic spanning software engineering, project management, information systems and artificial intelligence, where no single venue or database suffices. The review takes Monserrat et al. (2025) as the baseline for traditional software-project CSFs, then examines AI-focused literature to establish how AI applications relate to those factors and which conditions shape the outcome. The synthesis in Section 4 draws on 34 AI-focused studies, of which 32 are peer-reviewed and two remain preprints, together with one supporting software-project measurement study, 16 studies that establish the CSF and project-success baseline, and three methodological sources. Of the 34 AI-focused studies, 17 address software engineering or software projects directly, seven concern project management more generally, and ten come from general AI, human–computer-interaction or cross-sector organisational research. Table 6 in Appendix B lists every one of them with its focus, evidence scope and the section in which it is used.

Three qualifications follow from this design. First, the database search was executed and is reported in Section 3.2, but full-text eligibility was not assessed for every record retained at title screening. Second, the synthesis is narrative and conceptual: it maps plausible mechanisms onto an established factor taxonomy and does not establish causal effects. Third, several sources on explainability, decision support and organisational AI adoption (Sections 2.3, 4.4 and 4.5) come from general AI, human–computer-interaction or cross-sector management research rather than from software-project management. Their findings are applied here by extension. The assumption is that trust, transparency and readiness mechanisms observed in other decision-making and adoption contexts transfer plausibly to project settings – an assumption this review does not test.

3.2 Search Execution and Screening

Table 5 in Appendix A states the protocol in full. The search was executed on 6 September 2026 against OpenAlex, an open bibliographic index that aggregates records from the publishers relevant to this topic. An open index was chosen in preference to subscription databases such as Scopus or Web of Science so that the search can be replicated by any reader without institutional access. Each of the six strings was applied to titles and abstracts and restricted to the 2019–2026 window; Table 3 reports the yield of each string. The yield is uneven: S4 alone returns 142 of the 233 records, reflecting the recent volume of publication on generative AI and developer productivity relative to the other four synthesis categories.

Figure 1 summarises the resulting flow. The six strings returned 233 records. After four records with duplicate index identifiers and 35 further duplicate titles were removed, 194 unique records remained. Title-level screening against the inclusion and exclusion criteria excluded 37 records and retained 157. Quality assessment was pragmatic rather than formal: each included source was checked for transparency and relevance using the publication record, abstract and, where consulted, full text, considering whether it states its aim, adopts a suitable design, reports its data source, explains its analysis procedure, discusses limitations, and contributes to the research question. This was not a formal risk-of-bias assessment. Sources failing several checks

serve, if at all, as contextual background rather than as evidence for substantive claims.

Two further deviations from full systematic-review practice follow, beyond the qualifications already stated in Section 3.1. First, full-text eligibility was not assessed for all 157 retained records; the 34 AI-focused studies synthesised in Section 4 were drawn from the retained records and from reference-following. Selection prioritised peer-reviewed work that addresses the five synthesis categories directly. Second, only two of those 34 studies were retrieved by the six strings. To establish whether this reflects the strings or the topic, the six strings were validated against the 34 studies as a quasi-gold standard, following Zhang et al. (2011). They were then compared with four broader strings aligned to the synthesis categories (Table 2). The broader strings recover 15 of the 34 studies but return roughly 5,800 records, a volume beyond the screening capacity of a seminar paper; the protocol strings recover two studies from 233 records. The trade-off is structural: the relevant literature is dispersed across effort estimation, defect prediction, generative-AI productivity and organisational adoption, and rarely uses the phrase ‘software project’. Reference-following is therefore the principal route by which cross-disciplinary studies entered the synthesis. The review is accordingly a scoped rather than an exhaustive search, supplemented by snowballing after Wohlin (2014). The complete search log, every screening decision and the scripts used to reproduce them are published at <https://github.com/AhmedHedi0/ai-in-software-projects-csf>.

Table 2: Sensitivity of the search strings against the 34 AI-focused studies synthesised, following the quasi-gold-standard method of Zhang et al. (2011). OpenAlex, title and abstract, 2019–2026.

String set	Records	Known studies retrieved
Protocol strings S1–S6 (Table 3)	233	2 of 34
Broader category-aligned strings (pilot, not screened):		
effort and cost estimation with ML	469	4
generative AI and developer productivity	1,920	3
software defect prediction with ML	1,060	3
AI in project management, success and decisions	2,367	5
total	5,816	15 of 34

Table 3: Yield of each search string, OpenAlex, title and abstract, 2019–2026, executed 6 September 2026.

ID	Search string (title and abstract)	Records
S1	‘artificial intelligence’ AND ‘project success’ AND ‘software project’	15
S2	‘project management’ AND ‘critical success factors’ AND ‘artificial intelligence’	24
S3	‘machine learning’ AND ‘software project management’ AND success	19
S4	‘generative artificial intelligence’ AND ‘software development’ AND productivity	142
S5	‘artificial intelligence’ AND ‘software project planning’	5
S6	‘artificial intelligence’ AND ‘software project’ AND ‘risk management’	28
Total records identified		233
Unique records after duplicate removal		194

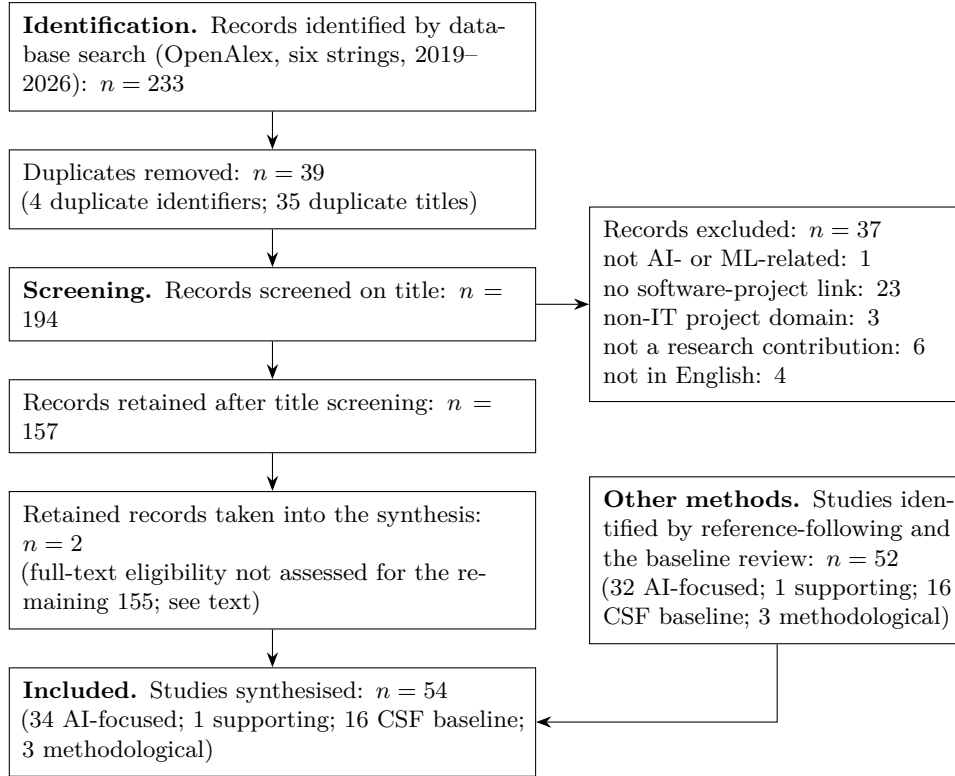


Figure 1: Flow of records through the review, following the structure of the PRISMA 2020 statement. Counts are those actually observed when the protocol was executed on 6 September 2026.

4 Evaluation and Synthesis

This section applies the five synthesis categories to the CSF baseline. For each category, it identifies the factors a class of AI applications may affect, the success dimensions involved, and the conditions under which the effect may fail to materialise.

4.1 Planning and Prediction

Planning is among the most firmly established CSFs in software projects. Monserrat et al. (2025) treat project planning, schedule management, economic management, risk management and the alignment of project objectives with the business model as separate factors within the Process and Project Management category. AI is relevant to all of them because it can process historical and real-time project data faster than manual analysis.

Predictive analytics and machine-learning models can support effort estimation, schedule forecasting and resource allocation. Two systematic reviews establish this base: Wen et al. (2012) synthesise machine-learning effort-estimation models across the earlier literature, and Ali and Gravino (2019) review 75 primary studies published between 1991 and 2017, finding artificial neural networks and support-vector machines to be the strongest-performing techniques. A bibliometric review reports comparable activity in AI-supported scheduling (Zahrour et al. 2023). Later work extends this to deep-learning and generative approaches (Uc-Cetina 2023; Mahmood et al. 2022; Tran et al. 2023). A model might, for instance, draw on past project size,

team composition, requirements volatility, defect rates, velocity data and dependency patterns to indicate an elevated risk of delay or cost overrun. Interview-based and secondary evidence suggests this may support decision quality where project managers treat AI forecasts as one input among several rather than as instructions to be followed (El Khatib and Al Falasi 2021).

The value of such tools is conditional. Forecasting systems require sufficient, relevant and high-quality data. Where historical data reflects poor measurement practices, superseded technologies, biased estimation behaviour or unrepresentative project contexts, predictions may be unreliable in ways that are not visible to the user. AI-supported planning therefore requires data governance and expert validation as prerequisites, not optional refinements. Predictive models also affect measurability: because they require structured historical data, their use may encourage organisations to formalise and record project indicators more consistently (Menezes Jr. et al. 2013), addressing one of the measurement gaps identified by Monserrat et al. (2025).

4.2 Delivery Performance

AI affects delivery performance through code generation, documentation support, test generation, code review, knowledge retrieval and defect prediction, the last of which has the longest empirical record (Giray et al. 2023; Jorayeva et al. 2022). These applications connect to development-related CSFs, in particular the specification of project requirements, coding standards, testing of implemented components, maintenance and evolution of software, and the choice of software-development life-cycle model.

Generative AI assistants can reduce the time spent on repetitive coding and documentation work. Drawing on a large sample of GitHub Copilot users, Ziegler et al. (2024) report that suggestion acceptance rates track developers’ perceived productivity. Because that study was conducted by researchers affiliated with the tool provider and reports tool-use indicators rather than project outcomes, it is not independent evidence of improved project-level performance. A pilot case study and two randomised controlled trials point in the same direction (Coutinho et al. 2024; Butler et al. 2025; Paradis et al. 2025). Assistants can help developers make sense of unfamiliar code, generate test cases, summarise technical documentation and offer suggestions during debugging. Such capabilities may strengthen the skills, knowledge and productivity of project participants, which are themselves Team and People factors in the baseline taxonomy; cross-sector reviews report comparable productivity effects outside software engineering (Al Naqbi et al. 2024).

However, delivery speed should not be equated with project success. Speed is a success criterion rather than a success factor: it measures an outcome instead of explaining it, and it can rise while the factors that sustain quality – testing discipline, maintainability and adherence to coding standards – weaken. Generated output can be incorrect, incomplete or inconsistent with project-specific standards and architecture, and should be reviewed and tested before it enters production code or project documentation. Testing, code review and accountable human approval therefore remain necessary safeguards; responsible use offsets part of the apparent productivity gain with the verification effort it requires (Simkute et al. 2024; Butler et al. 2025).

4.3 Decision-Making and Control

Project control involves tracking progress, comparing actual results against plans, monitoring risks and initiating corrective action. AI-enabled dashboards and anomaly-detection tools can support these activities by combining data from issue trackers, source-code repositories, test results, communication systems and delivery pipelines (Taboada et al. 2023).

Such tools relate to project planning, schedule management, quality management, risk management and change management in project planning. They may reveal warning signs that are difficult to detect manually, including unusual concentrations of defects, falling delivery velocity, rising rework, scope-growth patterns or unresolved dependencies.

Nevertheless, AI-supported control can create false confidence. Dashboards tend to emphasise what is easily measured while overlooking stakeholder dissatisfaction, interpersonal conflict, informal coordination or shifting strategic needs. Models produce false positives and false negatives, especially where data is incomplete or poorly integrated (El Khatib and Al Falasi 2021). Project managers must therefore interpret AI output in context and retain accountability for the resulting decisions. This qualification matches the wider finding that social and organisational conditions, not fully captured by instrumented data, shape development-project outcomes (McLeod and MacDonell 2011).

4.4 People and Stakeholders

The people dimension is central both to traditional CSF research and to AI adoption. Monserrat et al. (2025) identify participation, commitment, support, motivation, communication, skills and competencies, knowledge, training, experience, team characteristics, and confidence and security as software-project CSFs. These factors matter even more in distributed settings, where communication and coordination difficulties are among the most consequential risks (Niazi et al. 2016); trust and collaboration within project teams are themselves associated with project-management success (Bond-Barnard et al. 2018).

AI may help teams retrieve information, summarise discussions, draft communications, search project knowledge bases and identify recurring stakeholder concerns (Russo 2024; Taboada et al. 2023). It can improve communication efficiency and reduce information overload, and it may support requirements analysis by grouping and summarising user feedback, thereby touching the specification of project requirements and stakeholder management.

AI has, however, no evident substitute for the interpersonal work of leadership, negotiation, trust-building, conflict resolution or stakeholder sense-making: the reviewed literature offers little evidence that these activities can be automated, and they remain strongly dependent on human and contextual judgement. Automated summaries can omit nuance, misread context or expose confidential information. Research on explainable AI finds that users who cannot see how a recommendation was produced tend either to distrust it or to accept it uncritically (Rong et al. 2024; Rogha 2023); both responses degrade decision quality. Training, transparency, participation and clear responsibility structures are therefore essential accompaniments to AI adoption (Mohammad and Chirchir 2024; Chang et al. 2024).

4.5 Governance and Risk

Governance and risk are cross-cutting conditions rather than isolated technical issues. AI adoption affects organisational policies and rules, culture, structure, infrastructure, training, security and accountability, all of which fall within the Organisational and Team and People categories of the baseline taxonomy (Monserrat et al. 2025). Empirical work on AI adoption in general organisational settings points to similar conditions, identifying technology readiness, trust, transparency and perceived loss of control as decisive (Uren and Edwards 2023; Booyse and Scheepers 2023; Tariq et al. 2021).

Data quality is especially consequential. AI systems used for estimation, forecasting or monitoring depend on the validity, completeness, relevance and timeliness of project data (El Khatib and Al Falasi 2021). Poor data produces misleading recommendations and can amplify existing estimation or prioritisation biases rather than correcting them.

Explainability and accountability matter equally. The explainability literature argues, for decision support in general, that users must be able to see why a system produced a given recommendation (McDermid et al. 2021; Kostopoulos et al. 2024); the argument applies with particular force where recommendations affect budgets, schedules, staffing or risk decisions. Human oversight should ensure that decisions remain contestable, auditable and aligned with organisational objectives.

When generative tools process source code, requirements, business information, customer data or internal documentation, organisations should assess data-handling, confidentiality and licensing conditions before use; AI techniques have themselves been applied to security-risk assessment across the development life cycle (Olusanya et al. 2024). Organisations need policies that define permissible tools, approved data handling, validation requirements and responsibility for AI-assisted output.

Table 4 consolidates the mapping developed across Sections 4.1–4.5.

Table 4: Conceptual mapping of AI applications to the software-project CSF baseline of Monserrat et al. (2025), with associated success contributions and enabling conditions.

AI application	Traditional CSF(s) affected	Potential success contribution	AI-specific risks or enabling conditions
Predictive analytics and machine-learning forecasting	Project planning; schedule management; economic management; risk management	Better effort, cost, schedule and risk forecasts	Requires relevant, high-quality historical data, quality controls, domain validation and bias monitoring
Generative AI assistants	Skills, competencies and capacity; knowledge of participants; testing of implemented components; project documentation; maintenance and evolution	Faster coding, documentation, testing and knowledge access	Quality assurance, security, intellectual property, hallucination and overreliance risks
Intelligent monitoring and anomaly detection	Project planning; quality management; risk management; change management in project planning	Earlier identification of schedule, quality, scope or risk problems	False positives, incomplete data integration, weak interpretability and fixation on easily measured metrics
AI decision-support systems	Economic management; schedule management; risk management; stakeholder management	Faster prioritisation and resource-allocation support	Bias, opacity, unclear accountability and automation bias
AI-supported communication and requirements analysis	Communication; stakeholder management; specification of project requirements	Improved processing of feedback, issues and stakeholder signals	Privacy, trust, contextual misunderstanding and user-acceptance risks
AI governance and organisational readiness	Organisational culture; policies and rules of the organisation; training of participants; availability of tools and infrastructure	Supports safe and effective AI use across the project life cycle	Data governance, AI literacy, security, explainability, accountability and human oversight are prerequisites

Table 4 offers a conceptual synthesis, not evidence that AI universally causes project success. It pairs every claimed contribution with the condition on which that contribution depends, showing that AI can strengthen particular CSFs only where suitable technical and organisational conditions are in place. High-impact AI-supported project-management activities therefore generally require appropriate governance, data quality, participant competence and human oversight as cross-cutting enabling conditions, not separate concerns to address after adoption;

the relative weight of explainability and other safeguards depends on the application, the sensitivity of the data and the consequences of error. The evidence base underlying the AI column is heterogeneous in method, setting and rigour.

5 Discussion

5.1 Traditional Project-Management Perspective

The traditional perspective explains where AI is likely to offer the most direct benefits. Applications for estimation, scheduling, cost forecasting, quality monitoring and risk detection map closely onto the iron triangle and onto conventional control activities, the areas in which practitioners expect AI to exert its strongest influence (Fridgeirsson et al. 2021).

The traditional view is nevertheless incomplete, because it concentrates on delivery outcomes. It does not explain why one organisation obtains value from AI while another does not, nor does it address whether a technically successful project generates stakeholder satisfaction or strategic benefit (Ika 2009).

5.2 Critical-Success-Factor Perspective

The CSF perspective supplies that account. It asks not whether AI is useful in general, but which established success factors AI changes, and under what conditions. Table 4 answers the first half of that question, mapping each class of AI application onto the factors of Monserrat et al. (2025).

AI also introduces enabling conditions of its own: data quality, organisational policies, governance, training, tool availability, security, explainability and human oversight. Weakness in any one of them can prevent AI from improving the factors it targets. Based on this mapping, this review therefore proposes that AI is best understood as a conditional amplifier of established CSFs rather than as an independent one. Ahimbisibwe et al. (2015) advance a contingency logic for methodology choice that applies here too: the question is not which factors matter universally, but which configuration fits a given project and technological context.

5.3 Socio-Technical and Agile Perspective

A socio-technical perspective emphasises that software projects involve people, organisational structures, relationships and changing contexts, not only technical systems (McLeod and MacDonell 2011). The view matters for AI adoption because people must interpret, trust, challenge and absorb AI output into collaborative work before it can influence any outcome.

AI may support agile delivery by summarising information, assisting backlog refinement, generating test cases and surfacing emerging risks. Yet agile and socio-technical success also rests on team communication, customer collaboration, trust, autonomy, feedback, learning and adaptation (Chow and Cao 2008), none of which reduces to automated data processing. Evidence on the perceived causes of failure points in the same direction: most failure mechanisms lie in people and process rather than in tooling (Lehtinen et al. 2014).

The socio-technical view cautions against substituting AI recommendations for human judgement in contextual interpretation, ethical judgement, stakeholder engagement and conflict management.

5.4 Implications for IT Project Management

Several practical implications follow for managers of software development projects:

- Treat AI adoption as a change initiative with its own objectives, stakeholders and risks, not merely as a tool deployment.
- Begin with specific, high-value use cases such as estimation support, risk monitoring, test assistance, documentation or knowledge retrieval, and evaluate each before broadening the scope.
- Establish data-quality standards before relying on AI forecasts or recommendations, and require human validation together with defined ownership and accountability for high-impact output such as budget forecasts, staffing decisions, delivery commitments and risk escalations.
- Train project managers, developers and stakeholders in AI literacy, limitations, security, privacy and responsible use.
- Maintain project-level policies covering approved tools, source-code handling, confidential information and intellectual-property compliance.
- Evaluate AI against multidimensional success measures that include quality, delivery performance, user acceptance, stakeholder satisfaction and business value, rather than productivity alone.

5.5 Limitations

Four limitations qualify these findings. First, the synthesis in Section 4 is conceptual and maps plausible mechanisms onto an established taxonomy without demonstrating causal effects on project outcomes. Second, the AI-focused evidence base is heterogeneous in method and rigour and mostly examines tool-level rather than project-level outcomes (Treude and Storey 2025); two of the 34 AI-focused studies remain unrefereed preprints (Uc-Cetina 2023; Rogha 2023). Third, the search was scoped rather than exhaustive. The protocol strings retrieved only two of the 34 AI-focused studies, full-text eligibility was not assessed for the remaining retained records, and most studies entered through reference-following (Section 3.2); relevant work may therefore have been missed, and selection reflects the reviewer’s judgement more than a purely mechanical procedure. Fourth, the underlying CSF literature itself lacks consistent definitions and measurable indicators (Monserrat et al. 2025), so no AI effect on a given factor can yet be assessed precisely.

6 Conclusion

This paper examined how artificial intelligence influences critical success factors and project success in software development projects. The review indicates that AI does not operate as

an independent and universally positive success factor. Instead, it alters the effectiveness of established CSFs by supporting planning, prediction, monitoring, risk management, knowledge access, documentation, testing and selected decision-making activities; Table 4 pairs each activity with the contribution it may offer and the condition it depends on (RQ1).

AI adoption also introduces requirements and risks (RQ2). Beneficial outcomes depend on data quality, governance, organisational culture, AI literacy, explainability, privacy, security, intellectual-property management and human oversight; in their absence, AI may produce inaccurate forecasts, biased decisions, insecure code, privacy violations, weak accountability and misplaced trust in automated recommendations. This answers the primary research question: AI is best understood as a conditional amplifier of established critical success factors. Successful AI-enabled software development projects combine technical capability with strong governance, capable and engaged people, clear organisational policies and continued human judgement; they do not depend on AI adoption alone.

Monserrat et al. (2025) identify a further gap: the literature still lacks consistent definitions and measurable indicators for CSFs and for project success. Future research should undertake longitudinal and empirical studies linking particular AI applications to measurable project outcomes across time, cost, quality, stakeholder satisfaction, user acceptance and business value. It should also examine the conditions under which AI benefits are sustained or reversed across different organisational and project contexts.

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A Review Protocol

Table 5: Summary of the review protocol.

Element	Specification
Review type	Scoped systematic literature review following Kitchenham and Charters (2007), with supplementary snowballing after Wohlin (2014)
Research question	How does the adoption of artificial intelligence influence critical success factors and project success in software development projects?
Baseline taxonomy	The 50 CSFs in six categories consolidated by Monserrat et al. (2025)
Sources	OpenAlex (open bibliographic index), searched via its public REST API on 6 September 2026. An open index was used in preference to subscription databases so that the search is replicable without institutional access
Search strings	Six strings applied to titles and abstracts; stated verbatim with their yields in Table 3
Period	2019–2026 for AI-focused literature; earlier landmark studies retained for theoretical foundations
Inclusion criteria	Peer-reviewed English-language studies on AI, machine learning, generative AI or intelligent decision support in software or wider IT project contexts; studies on project success, performance, planning, estimation, risk, quality, delivery or stakeholders; foundational project-success and CSF studies used for theoretical context. Preprints are admitted only where no peer-reviewed version exists; two such sources are retained and are identified as preprints in the bibliography
Exclusion criteria	Purely technical AI-model studies without a project-management link; non-IT project studies without comparative relevance; opinion pieces without a transparent method; duplicate records
Screening and quality assessment	Title-level screening of 194 unique records against the criteria above, followed by quality assessment against six transparency criteria (Section 3.2)
Data-extraction fields	Bibliographic data; focus; AI technique; evidence scope (software, project management or general); synthesis category; section of use; peer-review status (Appendix B). Study design, sample and reported effect were not extracted, because full-text eligibility was not assessed for every retained record (Section 3.2)

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Element	Specification
Synthesis categories	Planning and prediction; delivery performance; decision-making and control; people and stakeholders; governance and risk
Reporting limitation	Full-text eligibility assessment was not carried out for all retained records, and the synthesis is presented as a conceptual mapping rather than as evidence of causal effects (Sections 3.1 and 3.2)

B AI-Focused Studies and Supporting Sources

Table 6: AI-focused studies and supporting sources synthesised in Section 4: 34 AI-focused studies and one supporting software-project measurement study (marked *), included to inform the measurement implications of AI-supported forecasting. Scope: SW = software engineering or software projects; PM = project management, not software-specific; GEN = general AI, human-computer-interaction or cross-sector organisational research. Entries marked † are preprints. Study design, sample and reported effect are deliberately omitted: full-text eligibility was not assessed for every record, and recording those fields would overstate what was verified.

Study	Focus	Scope	Used in
Fernandez Rodriguez and Magaña Martínez (2015)	AI applied to project success and factor identification	PM	2.3
Fridgeirsson et al. (2021)	Expert survey of expected AI influence across PMBOK knowledge areas	PM	2.3, 5.1
Mahdi et al. (2021)	Review of machine learning in software project management	SW	1, 2.3
Taboada et al. (2023)	Systematic review of AI-enabled project management	PM	1, 2.3, 4.3
Adamantiadou and Tsironis (2025)	Systematic review of AI applications in project management	PM	1
Müller et al. (2024)	Empirical overview and research agenda for AI in project management	PM	2.3

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Study	Focus	Scope	Used in
Mohammad and Chirchir (2024)	Challenges of integrating AI in software project planning	SW	4.4
Wen et al. (2012)	Systematic review of ML-based effort-estimation models	SW	4.1
Ali and Gravino (2019)	Systematic review of software effort prediction using ML	SW	4.1
Mahmood et al. (2022)	Performance evaluation of ML techniques for effort estimation	SW	4.1
Uc-Cetina (2023) [†]	Survey of recent ML advances in effort estimation	SW	4.1
Tran et al. (2023)	AI-based effort-estimation framework proposal	SW	4.1
Zahroun et al. (2023)	Bibliometric review of AI in project scheduling	PM	4.1
Menezes Jr. et al. (2013)*	Indicators for risk assessment in software projects (not AI-focused)	SW	4.1
Ziegler et al. (2024)	Copilot acceptance rates and perceived productivity (vendor-conducted)	SW	4.2
Coutinho et al. (2024)	Pilot case study of generative AI and developer productivity	SW	4.2
Butler et al. (2025)	Randomised controlled trial of generative AI coding tools	SW	4.2
Paradis et al. (2025)	Enterprise randomised controlled trial of AI on development speed	SW	4.2
Simkute et al. (2024)	Productivity loss in human–AI interaction	GEN	4.2
Al Naqbi et al. (2024)	Cross-sector review of generative AI and work productivity	GEN	4.2
Russo (2024)	Generative AI adoption in software engineering	SW	1, 2.3
Treude and Storey (2025)	Generative AI and empirical software-engineering method	SW	5.5
Akimova et al. (2021)	Survey of deep learning for software defect prediction	SW	2.3
Giray et al. (2023)	Use of deep learning in software defect prediction	SW	4.2
Jorayeva et al. (2022)	ML-based defect prediction for mobile applications	SW	4.2

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Study	Focus	Scope	Used in
El Khatib and Al Falasi (2021)	Effects of AI on project-management decision-making	PM	4.1, 4.3, 4.5
Rong et al. (2024)	Survey of user studies on model explanations	GEN	4.4
Rogha (2023) [†]	Review of explainable AI in AI-assisted decision-making	GEN	4.4
Kostopoulos et al. (2024)	Review of explainable AI decision-support systems	GEN	2.3, 4.5
McDermid et al. (2021)	Technical and ethical dimensions of AI explainability	GEN	4.5
Chang et al. (2024)	AI-driven technostress and adoption intention	GEN	4.4
Olusanya et al. (2024)	Neuro-fuzzy security-risk assessment across the SDLC	SW	4.5
Uren and Edwards (2023)	Technology readiness and organisational AI adoption	GEN	4.5
Booyse and Scheepers (2023)	Barriers to automated organisational decision-making	GEN	4.5
Tariq et al. (2021)	Driving forces and barriers to AI in operations	GEN	4.5